

文献数据库



88,343,822

篇文献 · 全文 80.5 TB

~20%

化学学科论文占比

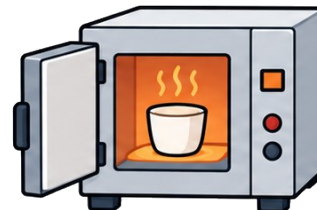
合成条件散落在正文段落中
尚未结构化



文献中的合成描述



结构化的合成知识



合成方案与实验装置

Could AI Dream of Science

大模型读懂文献里的合成经验，生成可执行的研究方案，交给自动化实验室执行与仿真

实验流程 Step 1-3



繁琐细致的实验步骤



① 称量后在玛瑙研钵中充分研磨混合



② 转移至刚玉坩埚



③ 分段热处理：350 / 1000 / 1300 °C

Experimental Guidance · 文献原文



“ The phosphors of **Sr_{4.75-5x-5y}Ca_{5x}Ba_{5y}(PO₄)₂(SiO₄):Eu_{0.252} + (0.01 ≤ x ≤ 1; 0.01 ≤ y ≤ 0.5)** were prepared: I) Sr_{4.75-5x}Ca_{5x}(PO₄)(SiO₄)_{0.5})₂:Eu_{0.252} + (0.01 ≤ x ≤ 0.95); and II) Sr_{4.75-5y}Ba_{5y}(PO₄)₂(SiO₄):Eu_{0.252} + (y = 0.01, 0.05, 0.1, 0.2, 0.5). The starting materials were **CaCO₃ (99.99%), SrCO₃ (99.99%), BaCO₃ (99.99%), NH₄H₂PO₄ (99.99%), SiO₂ (99.99%), Eu₂O₃ (99.99%)**. The stoichiometric chemicals **were weighed and thoroughly mixed in an agate mortar, then transferred to a corundum crucible and heated at 350 (in air), 1000 (in air) and 1300 °C (in reducing (h₂) atmosphere)** for 15 h, respectively. In the experiments no signals from Eu³⁺ were detected on the emission spectra, indicating the Eu³⁺ ions were successfully reduced to Eu²⁺ ions in the present experimental conditions. ”

一段话里的四类信息



组成与掺杂



起始原料



操作：称量 · 研磨 · 装料



热处理程序与气氛

简短且复杂的高密度知识信息 → 由大模型抽取为结构化合成方案（组成 · 原料 · 操作 · 条件）